B4M36SAN BE4M36SAN Seminar/Tutorial

Alikhan Anuarbekov

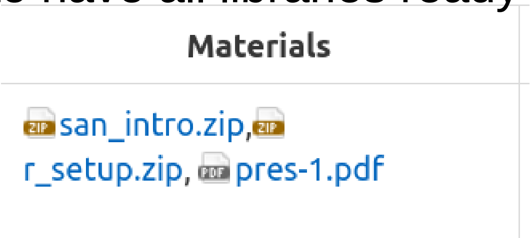
Ali

anuarali@fel.cvut.cz

- For those who don’t see the course in the left part:

  https://cw.fel.cvut.cz/wiki/courses/b4m36san/start

- If you want to have all libraries ready for labs, download **r_setup.zip**



- Today will be first english lecture after seminar (18:00 – 19:30), prof. Klema will collect you after seminar
  - You should have prepared by reading the first lecture.
  - If you didn’t get an email, mention it to prof. Klema!

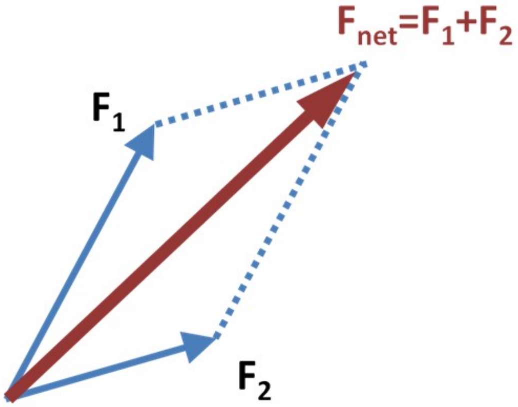
- Today is attendance control, you should mark the presence

Also, we collect feedback about prerequisites. Please answer the questionnaire:

https://forms.gle/5W4T6M9xMjHPaEybA

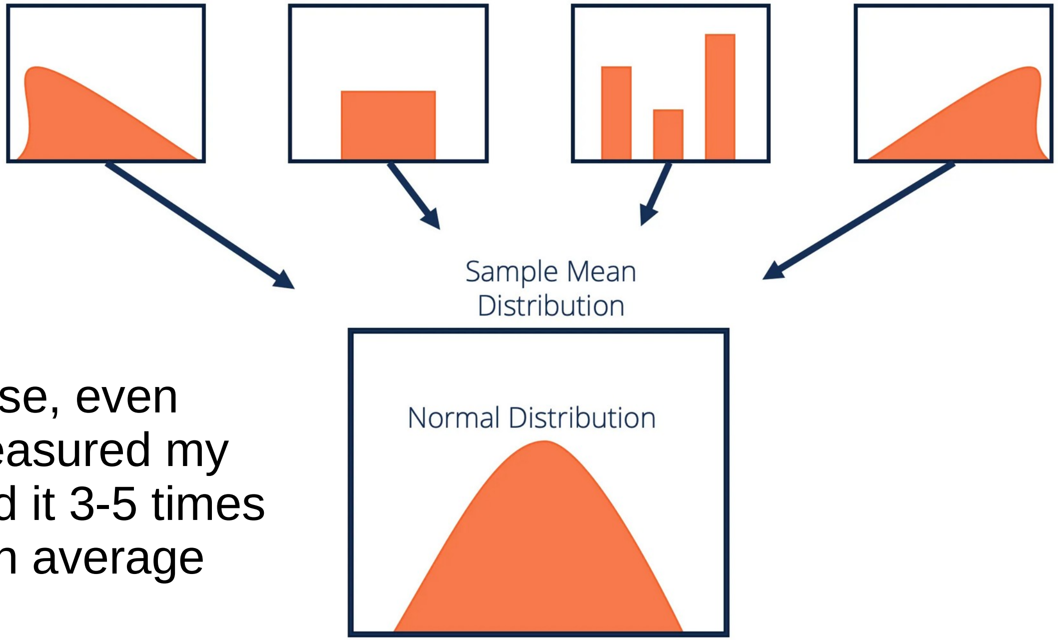
Why Gaussian normal distribution?

- Why we use a Gaussian normal? What if there is another distribution?
- Maybe, but typically when you have multiple factors/causes, they are typically summed



Example: Physics
sum of multiple forces

- Then, when we compute their mean (sum / number), it is mathematically Gaussian!
  (Central limit theorem)



- **Consequence** – to be more precise, even in previous test they didn't just measured my biomolecules level, they measured it 3-5 times from big blood sample and took an average

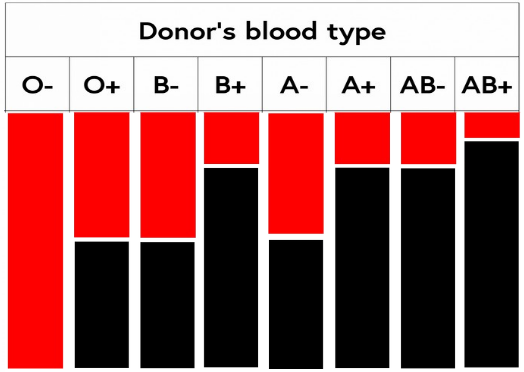
A little more about basic statistics

- But how do people actually found different types of distribution?
- Again – randomness comes from hiding certain variables

Example (2D space -> 1D distribution):

You need to performed a blood transfusion, but you don’t have time to measure patient blood type. You only know donor’s blood type. Then you have an uncertainty about transfusion success

	Donor's blood type							
Recipient's blood type	O-	O+	B-	B+	A-	A+	AB-	AB+
AB+	✓	✓	✓	✓	✓	✓	✓	✓
AB-	✓		✓		✓		✓	
A+	✓	✓			✓	✓		
A-	✓				✓			
B+	✓	✓	✓	✓				
B-	✓		✓					
O+	✓	✓						
O-	✓							

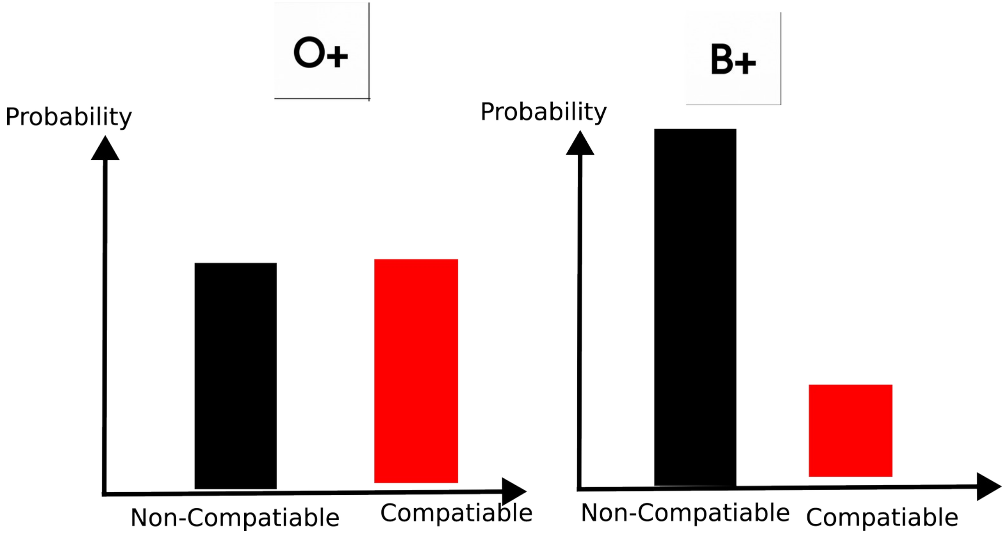


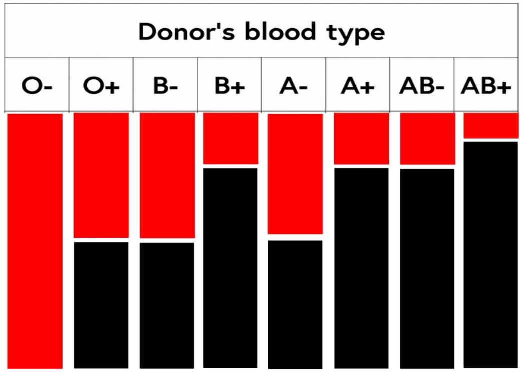
Practical example:

Human body: 3.2 billion DNA bases, 30 trillion cells,...

Distribution from histogram

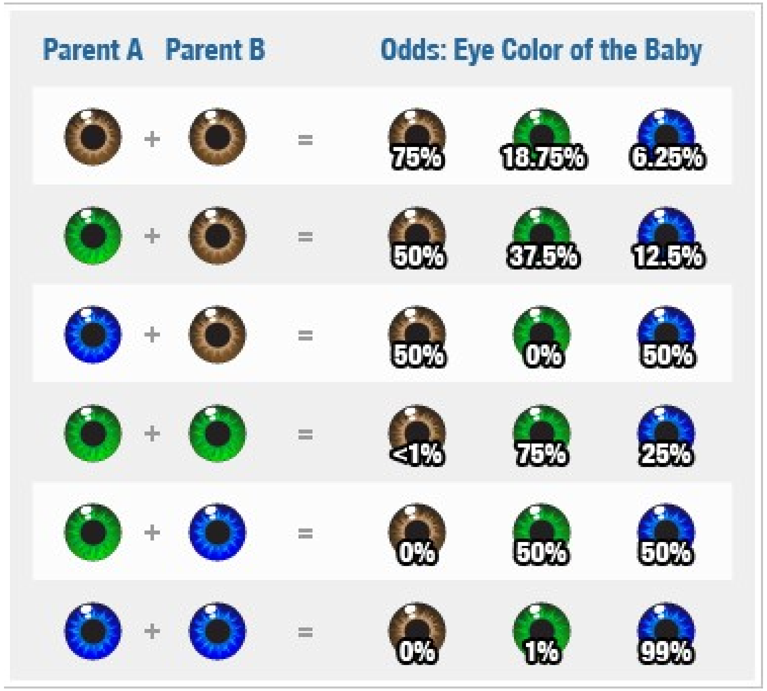
- And here scientists actually found out that “hiding” a variable produces certain shapes based on type of the ***<u>output</u>*** (e.g. response/dependent) variable
- For example, in previous example the ***dependent*** variable is ***“Is it compatible?”***
- This is finite (**discrete**) variable and it has 2 values = ***Compatible*** and ***Non-compatible***
- Known variables (not hidden) that we use to predict the dependent outcome are **independent** variables (e.g. we assume that they don’t depend to each other)



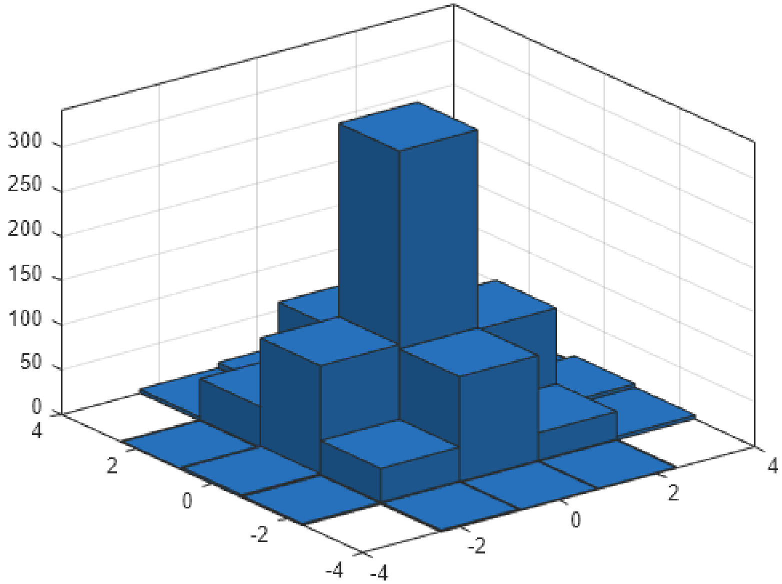


Histogram = count the number of outcomes (dep. variable values) with different hidden variable values that correspond to this particular independent value

Control question:

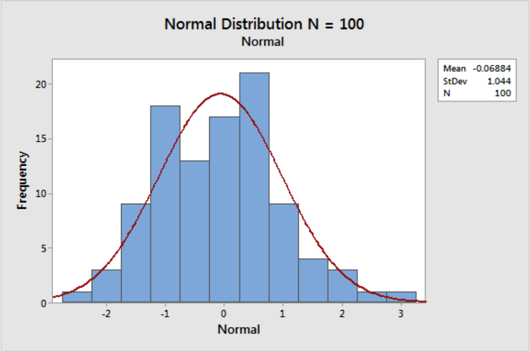


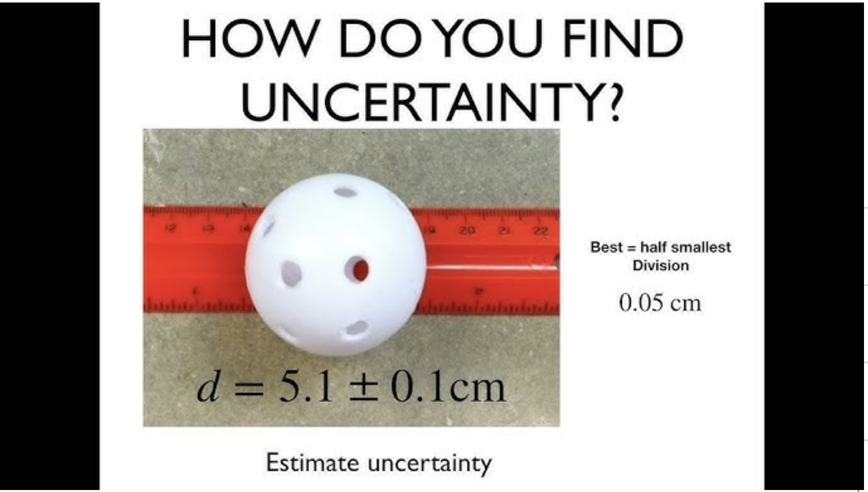
- What are independent variables?
- Are they discrete or continuous? How many values?
- What is the dependent variable?
- How will histogram look like for some points?

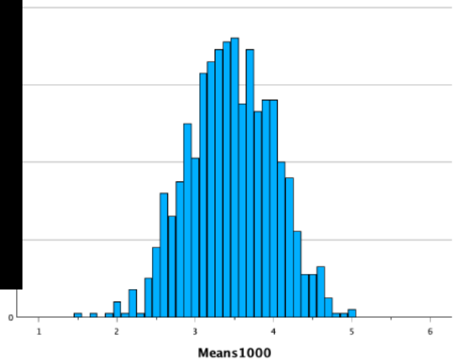


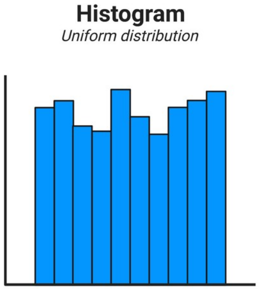
Distribution from histogram for continuous variables

- Same principle, but the dependent and independent variables can be ***continuous***
- But in practice we still *discretise* them for simplicity



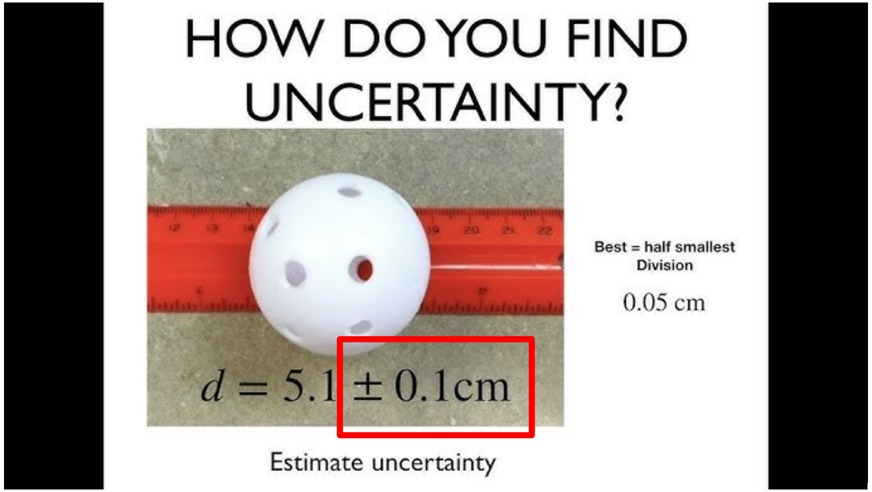


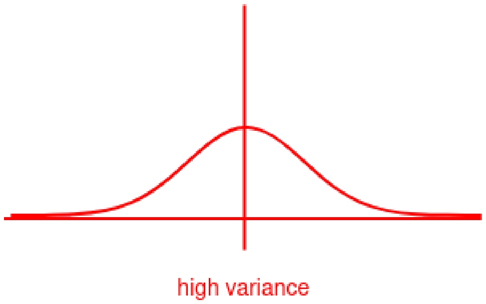


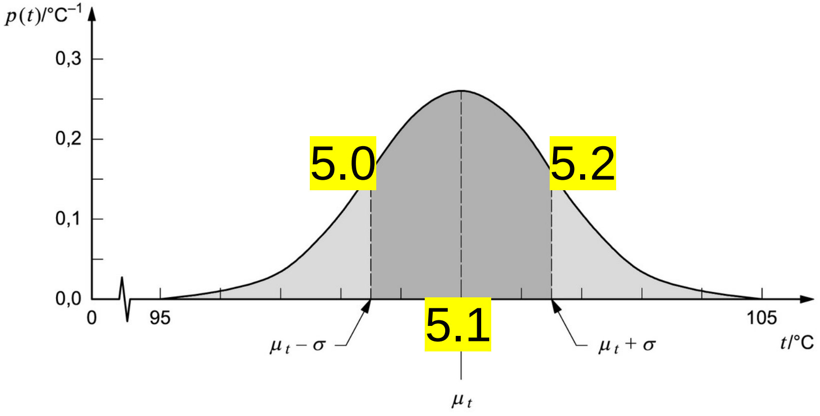


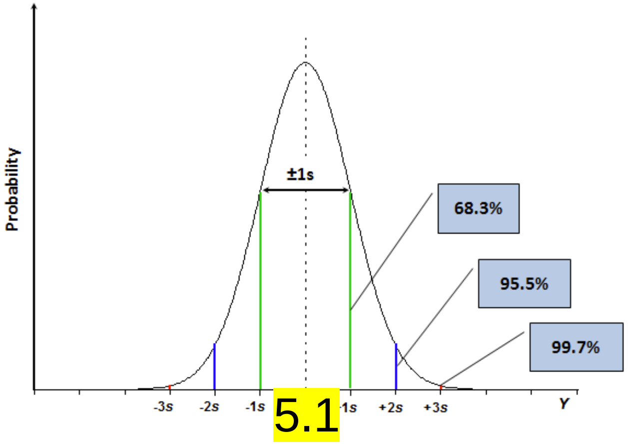
Y = dependent variable = real length
X = independent variable = number on ruler
other factors = error on factory that ruler was produced from, limitation of human perception, etc.

Variance is the quality of distribution for continuous variable

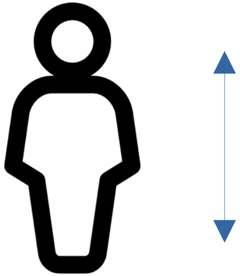




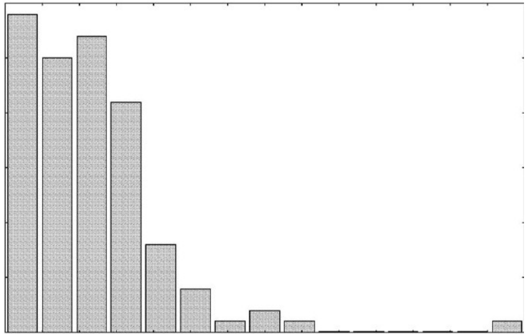




Previously:



H = Height,
Random variable



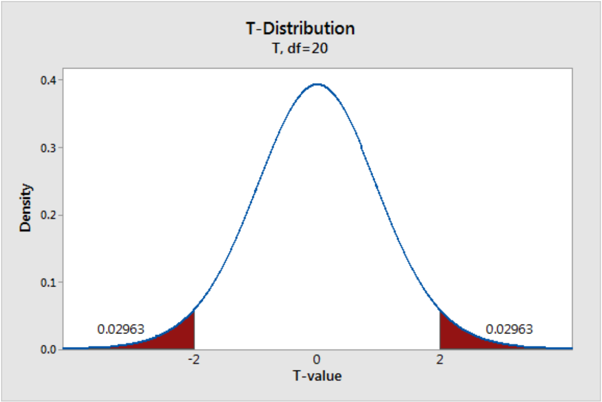
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Model as distribution

Test some hypothesis

$$\overline{H} = \frac{1}{N} \sum H_i \in N\left(\mu, \frac{1}{N}\sigma^2\right)$$

$$\frac{\overline{H} - mu}{\frac{1}{\sqrt{N}} S_H} \in t - student\left(df = N - 1\right)$$

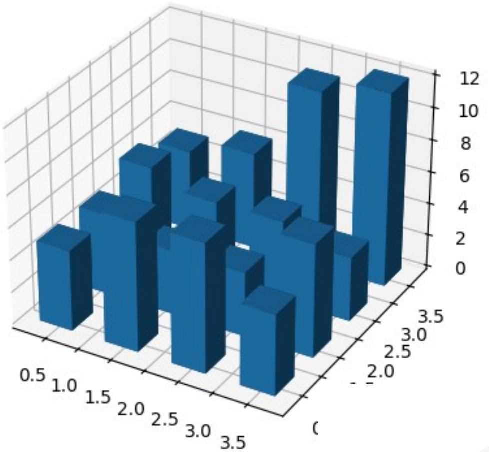


Problem of distribution-only modeling for more variables

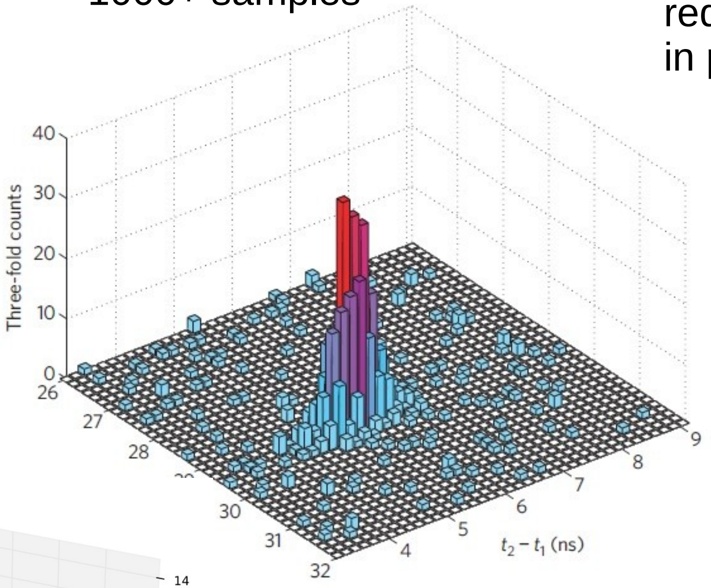
CURSE OF DIMENSIONALITY!

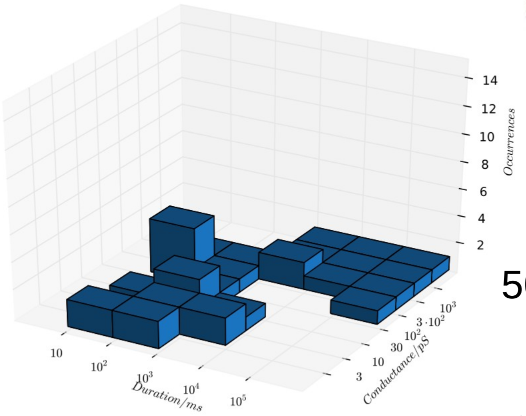
1000+ samples

1D Gaussian normal required just > 30 samples in practice



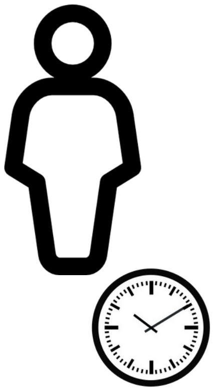
300 samples



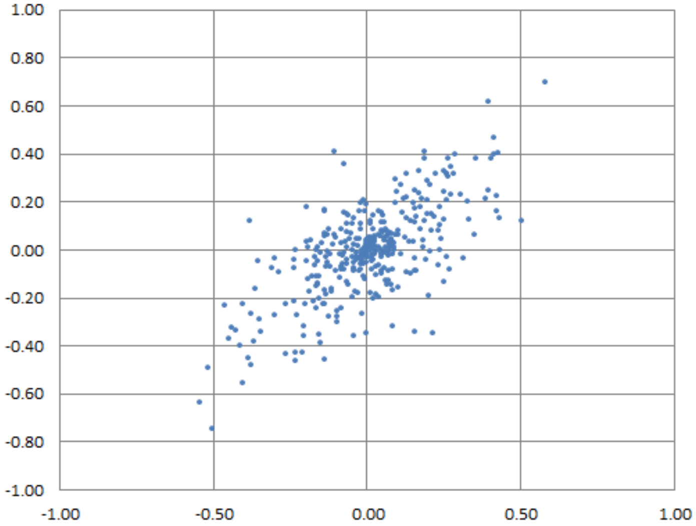


50 samples

Now:



Y = Height, X = Age
Both are random variables



m = number of points

Model as lin.regression

$$Y = \beta_0 + \beta_1 \cdot X + \epsilon,$$
$$\epsilon \in N(0, \sigma^2)$$

Test some hypothesis

H0: X and Y are not related, e.g. $\beta_1 = 0$

HA: X and Y have some relationship ($\beta_1 \neq 0$)

$$RSS = \sum (y_i - x_i \cdot \beta_1 - \beta_0)^2 \rightarrow \quad local\ optimum(estimate): \quad \hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Alternative: I assume that I know the TYPE (model) of formula between Y and X1/X2/….
Then I can derive relations for every human independently

For example: Human age = V is increasing -----> this causes the H=height to grow
Or other way around: H=height is increasing ----> this means that human ages

Assume: H = f(V) or f(V, H) = 0 most easiest function is linear:

$$H=\beta_0+\beta_1\cdot V \qquad \text{or} \qquad \alpha_0+\alpha_1\cdot H+\alpha_2\cdot V=0$$

Linearity assumption

But there are other factors, reality isn't that primitive:
* How often does this human eats a day? Note: discrete(categorical) variable!
* Does he/she have some disease that makes it age faster? Note: this too = Yes/No

Lets just say that all the other factors are secondary and approximate them:
(Central Limit Theorem)

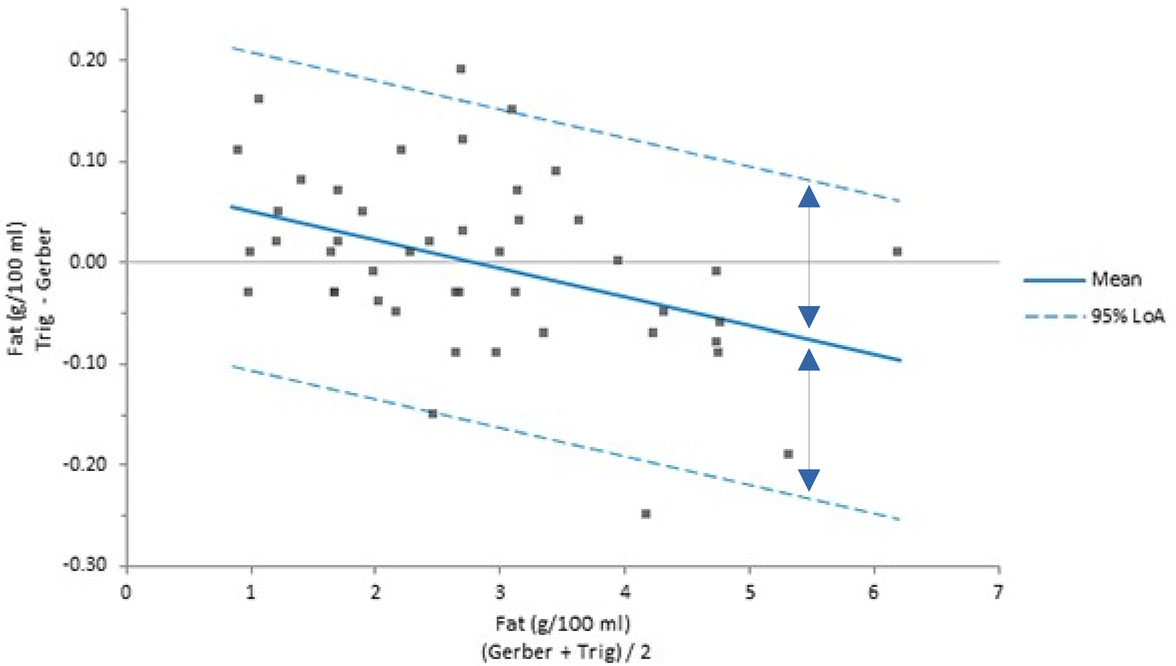
$$hidden_1+hidden_2+....\approx\epsilon, \quad \epsilon\in N\left(\underline{0},\sigma^2\right)\left(\text{if mean is not zero, just move it to intercept}\right)$$

$$H=\beta_0+\beta_1\cdot V+\epsilon, \qquad \epsilon\in N\left(0,\sigma^2\right)$$

Normality of error

$$H = \beta_0 + \beta_1 \cdot V + \epsilon, \qquad \epsilon \in N(0, \sigma^2)$$

We say that secondary factors are limited by some effect power = σ



Since these effects shouldn't change the slope(vertical tilt / incline ↕), it should have same variance across entire axis = Homoscedasticity assumption

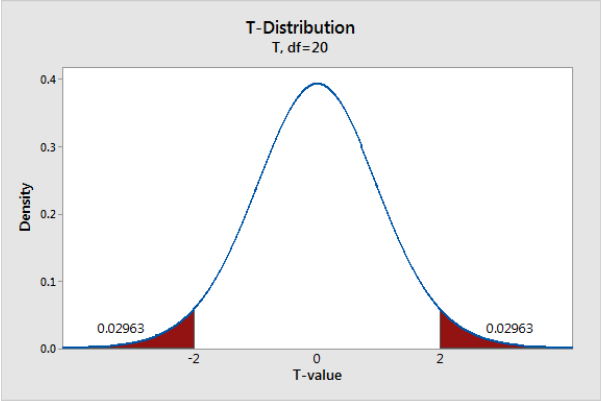
Given the data, we can find the best estimate explicitly:

$$RSS=\sum(y_i-x_i\cdot\beta_1-\beta_0)^2 \quad \rightarrow \quad local\ optimum(estimate): \quad \hat{\beta}_1=\frac{\sum(x_i-\overline{x})(y_i-\overline{y})}{\sum(x_i-\overline{x})^2}$$

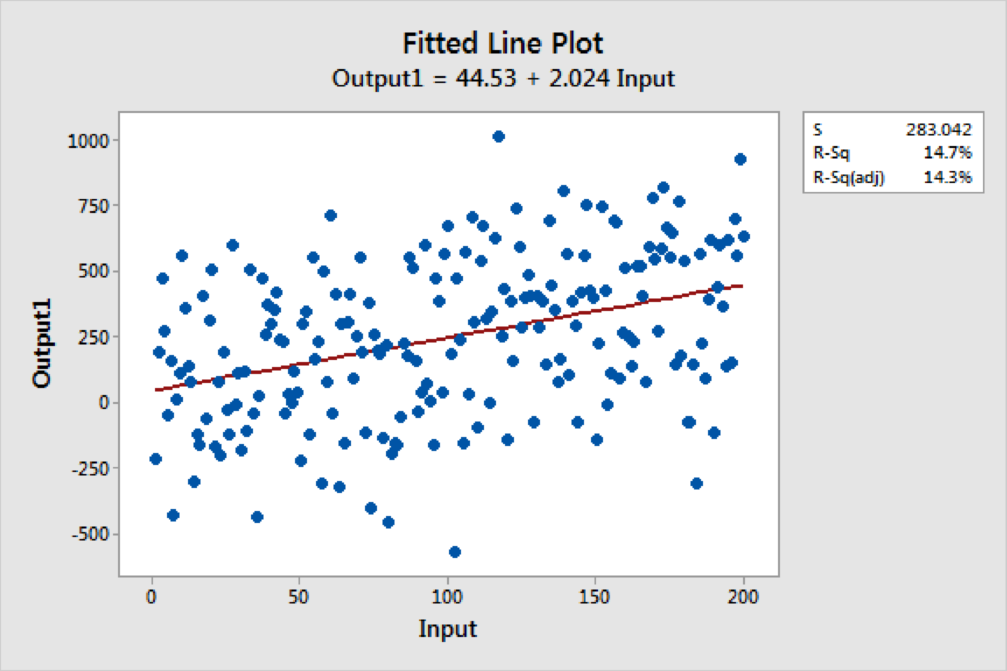
$$\frac{\hat{\beta}_1-\beta_1}{S_{\beta_1}}\in t-student(df=m-2)$$

H0: H and V are not related, e.g. $\beta_1=0$

HA: H and V have some relationship ($\beta_1\neq 0$)



Problem 1:



$$\epsilon \in N(0, \sigma^2)$$

If variance is too large, then our model is bad….

$$RSE = S_\epsilon = \sqrt{RSS/(m-2)}$$

But how to know when it is too large?

Solution:

$$RSE = S_\epsilon = \sqrt{RSS/(m-2)}$$
$$RSE \propto RSS$$

Two models, who has less variance?

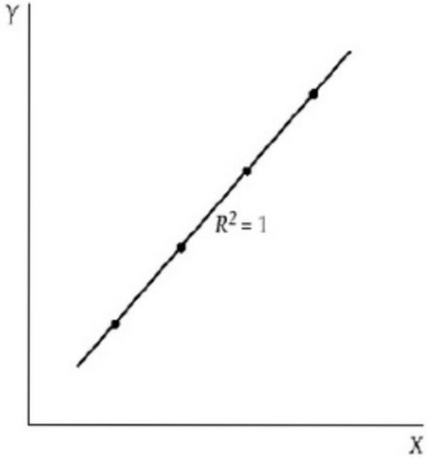
$$with\ \beta_1 \neq 0 : RSE \propto RSS$$

$$with\ \beta_1 = 0 : RSE_{null} \propto RSS_{null} = TSS$$

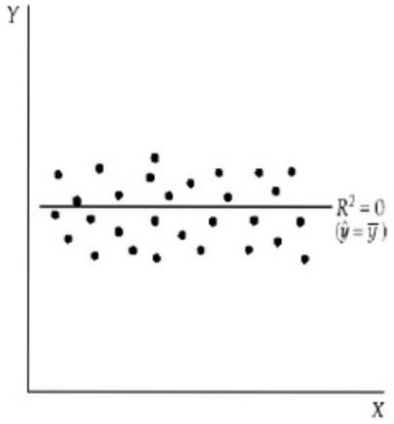
$$R^2 = 1 - \frac{RSS}{TSS}$$

Or if more precise method is needed:

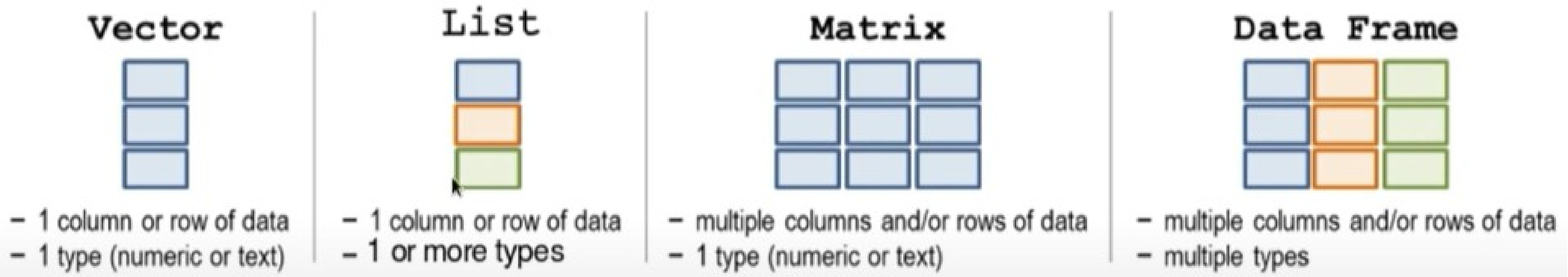
$$\frac{1}{m-2}\frac{RSS}{TSS} \in F - stat(order = 1, df = m-2)$$



If R^2 = 1 the total deviation in Y from its mean is explained by the equation.



If R^2 = 0 the regression equation does not account for any of the variation of Y from its mean.



	dma	day	amount_of_deaths	new_deaths	weekly_deaths	gtrends
66	1	2021-05-02	66247	47	491	62
67	1	2021-05-09	66690	41	443	62
68	1	2021-05-16	67030	43	340	55
7481	2	2020-02-02	0	0	0	68
7482	2	2020-02-09	0	0	0	66
7483	2	2020-02-16	0	0	0	68
7484	2	2020-02-23	0	0	0	65
7485	2	2020-03-01	0	0	0	66
7486	2	2020-03-08	0	0	0	66
7487	2	2020-03-15	1	0	1	68
7488	2	2020-03-22	12	3	11	77
7489	2	2020-03-29	56	6	44	79
7490	2	2020-04-05	184	16	128	100
7491	2	2020-04-12	397	38	213	96
7492	2	2020-04-19	777	25	380	77
7493	2	2020-04-26	1170	21	393	77
7494	2	2020-05-03	1559	20	389	73
7495	2	2020-05-10	1945	21	386	69

Showing 66 to 84 of 14,212 entries, 6 total columns

<----- **View**(**myDataFrame**)

myDataFrame$new_deaths = access column as vector

$$H = \beta_0 + \beta_1 \cdot V + \beta_2 \cdot K + \epsilon, \qquad \epsilon \in N(0, \sigma^2)$$

can be denoted shortly as (R notation): H ~ V + K

```
lm.fit <- lm(medv ~ lstat, data = Boston)
```

Also a useful notation: **H ~ .** (H = regression of all other variables)

If you are not sure about something about function/data:

help(lm) **help(Boston)**

?lm **?Boston**